

Solvers for Hierarchical PG–NB–GP: From Joint CG to Direct Schur

Scratchpad

1 Setup

Hierarchical additive GP:

$$f_i = g(t_i) + h_{\ell_i}(t_i), \quad g \sim \mathcal{GP}(0, k_g), \quad h_\ell \sim \mathcal{GP}(0, k_h), \quad \ell = 1, \dots, L.$$

Negative binomial likelihood with Pólya-Gamma augmentation gives the conditional:

$$p(f \mid \omega, y) \propto \exp\left(-\frac{1}{2}f^\top \Delta f + \kappa^\top f\right) \cdot \exp\left(-\frac{1}{2}f^\top K^{-1}f\right),$$

where $\Delta = \text{diag}(\omega_i)$, $\kappa_i = \frac{1}{2}(y_i - r)$, and K is the joint prior covariance of $f = g + h$.

2 Feature-space representation

Using the spectral approximation with shared grid $\{\xi_k\}_{k=1}^m$:

$$g(t) = \sum_k w_{g,k} a_{g,k} e^{2\pi i \xi_k \cdot t}, \quad h_\ell(t) = \sum_k w_{h,k} a_{h,k}^{(\ell)} e^{2\pi i \xi_k \cdot t},$$

where $w_{g,k} = S_g(\xi_k)^{1/2}$, $w_{h,k} = S_h(\xi_k)^{1/2}$ are spectral amplitude weights.

Define:

- $D_g = \text{diag}(w_{g,k}^2)$, $D_h = \text{diag}(w_{h,k}^2)$, and their square roots.
- $F \in \mathbb{C}^{n \times m}$: the NUFFT matrix, $F_{ik} = e^{2\pi i \xi_k t_i}$.
- $F_\ell \in \mathbb{C}^{n_\ell \times m}$: restriction to site ℓ .
- $T_\ell = F_\ell^* \Delta_\ell F_\ell$ (Toeplitz via NUFFT), $T_{\text{all}} = \sum_\ell T_\ell$.

3 The joint feature-space system

The posterior precision in symmetrized feature space is the $(1 + L)m \times (1 + L)m$ block matrix:

$$M = \begin{pmatrix} A & B_1 & \cdots & B_L \\ B_1^* & D_1^{(h)} & & \\ \vdots & & \ddots & \\ B_L^* & & & D_L^{(h)} \end{pmatrix}$$

where:

$$A = I_m + D_g^{1/2} T_{\text{all}} D_g^{1/2} \quad (m \times m, \text{global}) \quad (1)$$

$$D_\ell^{(h)} = I_m + D_h^{1/2} T_\ell D_h^{1/2} \quad (m \times m, \text{per-site}) \quad (2)$$

$$B_\ell = D_g^{1/2} T_\ell D_h^{1/2} \quad (m \times m, \text{cross-term}) \quad (3)$$

The Schur complement for g :

$$S = A - \sum_{\ell=1}^L B_\ell (D_\ell^{(h)})^{-1} B_\ell^* \quad (m \times m). \quad (4)$$

4 Overview of approaches

We describe four solver approaches, ordered by development. All share the same EM outer loop: an E-step that updates $\tilde{a}_g, \tilde{a}_h^{(\ell)}, \sigma_{\text{diag}}$, and an M-step that updates kernel hyperparameters via $\partial \mathcal{L} / \partial \theta$.

The key distinctions are:

1. How the E-step posterior mean is obtained (CG on $(1+L)m$ system, block CD, or Schur elimination).
2. How σ_{diag} is estimated.
3. How the M-step trace term $\text{tr}(M^{-1} \partial M / \partial \theta)$ is computed (observation-space Hutchinson, feature-space Hutchinson, or exact).

5 Approach A: Joint CG solver (baseline)

E-step. Solve the full $(1+L)m$ -dimensional system $M \tilde{a} = \text{rhs}$ with conjugate gradients. Each CG iteration applies the block matvec via Toeplitz embeddings: $O((1+L)m \log m)$ per iteration.

sigma_diag. Observation-space Hutchinson: for each probe $z \in \mathbb{R}^n$, solve $(K^{-1} + \Delta)^{-1} z$ via CG on the full system, transforming between observation and feature space at each CG step. Requires $2(1+L)$ NUFFT per CG iteration.

M-step. Observation-space Hutchinson probes: for each probe, apply $\partial K / \partial \theta$ (Toeplitz matvec in feature space, then NUFFT back), and compute $v^\top (\partial K / \partial \theta) v$ where $v = M^{-1} z$ from an additional CG solve. Requires $\sim 2(1+L)$ NUFFT per probe per CG iteration.

Cost per outer iteration. Dominated by NUFFT inside CG loops. Typical CG counts: $\sim 25,000$ – $36,000$ total CG iterations, each touching $O(n)$ NUFFT.

Benchmark. 1331s total, 88.7s/iter (mosquito dataset, $n=24,513$, $L=94$, $m \approx 25$).

6 Approach B: Block coordinate descent + Schur M-step

Key idea. Exploit the arrow structure of M : the h -blocks are decoupled given g . Alternate between updating g (fixing all h_ℓ) and updating each h_ℓ (fixing g), each via CG on an $m \times m$ system.

E-step. Each CD sweep:

- **g-step:** solve $A\tilde{a}_g = D_g^{1/2}F^*(\kappa - \Delta h_{\text{combined}})$ via CG. Cost: 1 NUFFT (type 1) for RHS, CG iterations using Toeplitz matvecs, 1 NUFFT (type 2) to recover g in observation space.
- **h-step:** for each ℓ , solve $D_\ell^{(h)}\tilde{a}_h^{(\ell)} = D_h^{1/2}F_\ell^*(\kappa_\ell - \Delta_\ell g_\ell)$ via CG. Cost: L NUFFTs (type 1) + L NUFFTs (type 2).

Per sweep: $2(1+L)$ NUFFTs. With 5 sweeps: $10(1+L)$ NUFFTs.

sigma_diag. Independent (uncoupled) Hutchinson: solve g and each h_ℓ component separately via CG, batching probes. Cost: $2(1+L)$ NUFFTs (amortized over probes).

M-step. Feature-space Schur: form S from the Toeplitz blocks (materializing T_ℓ via m basis-vector matvecs), then compute $\text{tr}(S^{-1}\partial A/\partial\theta_g)$ and analogous θ_h terms using Hutchinson probes in $(1+L)m$ -dimensional feature space. No observation-space NUFFTs needed.

Cost per outer iteration. E-step: $3 \times [10(1+L) + 2(1+L)] = 36(1+L) \approx 3420$ NUFFTs. M-step: ~ 4000 feature-space CG iterations (Toeplitz only). No M-step NUFFTs.

Benchmark. 649s total, 43.3s/iter. Speedup: **2.0** $\times$ over joint.

Limitation. CD is an approximation: the g/h blocks are not jointly optimal at any finite number of sweeps. Hyperparameters (especially σ_g^2) can drift relative to the joint solve, though predictions remain accurate.

7 Approach C: Schur CG (fully iterative)

Key idea. Eliminate h analytically via the Schur complement (??), reducing the E-step to an $m \times m$ CG solve for $\tilde{a}_g$. Then recover each $\tilde{a}_h^{(\ell)}$ by backsubstitution. This is *exact* (no CD approximation) and stays entirely in feature space after the initial NUFFT transform.

E-step.

1. Transform κ to feature space: $\hat{\kappa}_g = D_g^{1/2}F^*\kappa$, $\hat{\kappa}_{h,\ell} = D_h^{1/2}F_\ell^*\kappa_\ell$. Cost: $1+L$ NUFFTs (type 1).
2. Compute effective RHS: $b_g^{\text{eff}} = \hat{\kappa}_g - \sum_\ell B_\ell (D_\ell^{(h)})^{-1} \hat{\kappa}_{h,\ell}$. Each $(D_\ell^{(h)})^{-1}$ application uses CG with Toeplitz matvecs: $O(m \log m)$ per CG step.
3. Solve $S\tilde{a}_g = b_g^{\text{eff}}$ via CG. Each CG matvec applies $S = A - \sum_\ell B_\ell (D_\ell^{(h)})^{-1} B_\ell^*$, requiring L inner CG solves of size m per outer CG step. All Toeplitz, no NUFFTs.
4. Recover $\tilde{a}_h^{(\ell)} = (D_\ell^{(h)})^{-1}(\hat{\kappa}_{h,\ell} - B_\ell^*\tilde{a}_g)$.
5. Transform back: $g = F D_g^{1/2} \tilde{a}_g$ etc. Cost: $1+L$ NUFFTs (type 2).

Total: $2(1+L)$ NUFFTs per E-step inner iteration.

sigma_diag. Same as Approach B: independent batched Hutchinson with $2(1+L)$ NUFFTs.

M-step. Feature-space Hutchinson probes with the Schur matvec (same as Approach B). No NUFFTs.

Cost per outer iteration. E-step: $3 \times [2(1+L) + 2(1+L)] = 12(1+L) \approx 1140$ NUFFTs ($3 \times$ fewer than Block CD). But the Schur CG matvec is expensive: each application requires L inner CG solves, so the Toeplitz matvec count is high.

Benchmark. 619s total, 41.3s/iter. Speedup: **2.2** $\times$ over joint.

Tradeoff. Fewer NUFFTs than Block CD, but the nested CG (inner solves inside outer CG) has high Toeplitz overhead. Generalizes naturally to $d \geq 2$ since no matrix factorization is needed.

8 Approach D: Direct Schur (dense factorization)

Key idea. Since $m \approx 25$ (for $d=1$ with typical lengthscales), explicitly form T_ℓ , A , $D_\ell^{(h)}$, B_ℓ , and S as dense $m \times m$ matrices, then use Cholesky factorization for all solves. This eliminates CG entirely from the E-step and M-step.

Block construction. Form each $T_\ell \in \mathbb{C}^{m \times m}$ by applying the Toeplitz embedding to m standard basis vectors: cost $O(m^2 \log m)$ per site. Then:

$$A = I + D_g^{1/2} T_{\text{all}} D_g^{1/2}, \quad \text{Cholesky: } O(m^3) \quad (5)$$

$$D_\ell^{(h)} = I + D_h^{1/2} T_\ell D_h^{1/2}, \quad \text{Cholesky: } O(Lm^3) \quad (6)$$

$$B_\ell = D_g^{1/2} T_\ell D_h^{1/2}, \quad O(Lm^2) \quad (7)$$

$$S = A - \sum_\ell B_\ell (D_\ell^{(h)})^{-1} B_\ell^*, \quad \text{Cholesky: } O(Lm^3) \quad (8)$$

Total: $O(Lm^3)$. With $L=94$, $m=25$: $\sim 1.5 \times 10^6$ ops. **Negligible** (< 1 ms).

E-step.

1. Transform κ to feature space: $1+L$ NUFFTs.
2. Compute effective RHS using pre-factorized $(D_\ell^{(h)})^{-1}$: direct backsolves, $O(Lm^2)$.
3. Solve $S \tilde{a}_g = b_g^{\text{eff}}$: Cholesky backsolve, $O(m^2)$.
4. Recover $\tilde{a}_h^{(\ell)}$: direct backsolves, $O(Lm^2)$.
5. Transform back: $1+L$ NUFFTs.

Total: $2(1+L)$ NUFFTs + $O(Lm^2)$ algebra per inner iteration.

sigma_diag. **Exact** Schur diagonal: compute $[M^{-1}]_{ii}$ for each observation using the Schur block inversion formulas and pre-factorized matrices. Can also use Hutchinson with direct solves (no CG). Cost: $2(1+L)$ NUFFTs.

M-step. Exact trace — no Hutchinson probes needed. Since S^{-1} and $\partial A/\partial\theta$ are available as dense $m \times m$ matrices:

$$\text{tr}(M^{-1} \partial M / \partial \theta_g) = \text{tr}(S^{-1} \partial A / \partial \theta_g), \quad O(m^2) \quad (9)$$

$$\text{tr}(M^{-1} \partial M / \partial \theta_h) = \sum_{\ell} [\text{tr}(\cdots) + 2\Re \text{tr}(\cdots)], \quad O(Lm^2) \quad (10)$$

Zero NUFFTs, zero CG, zero stochastic estimation.

Cost per outer iteration. E-step: $3 \times 2(1+L) = 570$ NUFFTs + $O(Lm^2)$ dense algebra. M-step: $O(Lm^2)$ exact. **No CG anywhere.**

Benchmark. 159s total, 10.6s/iter. Speedup: **8.4** $\times$ over joint.

Limitation. Requires m to be small enough for dense $m \times m$ factorization ($m \lesssim 200$). This holds for $d=1$ with typical lengthscales, but for $d \geq 2$ the spectral grid grows as m^d , making dense factorization infeasible. In that regime, Approach C (Schur CG) is preferred.

9 Comparison

	A: Joint	B: Block CD	C: Schur CG	D: Direct
E-step solve	CG on $(1+L)m$	CD sweeps, CG on m	Schur CG on m	Cholesky on m
E-step NUFFTs/inner	many (in CG)	$10(1+L)$	$2(1+L)$	$2(1+L)$
sigma_diag	obs-space CG	indep. CG	indep. CG	exact / direct
M-step	obs-space Hutch.	feat-space Hutch.	feat-space Hutch.	exact trace
M-step NUFFTs	many	0	0	0
CG required?	yes	yes	yes (nested)	no
Exact joint post.?	yes	no (CD approx.)	yes	yes
Scales to $d \geq 2$?	yes	yes	yes	no (m^d)
Time/iter (s)	88.7	43.3	41.3	10.6
Total time (s)	1331	649	619	159
Speedup vs A	1.0 $\times$	2.0 $\times$	2.2 $\times$	8.4$\times$
Final MAE	—	27.4	27.3	27.4

Table 1: Comparison on mosquito dataset ($n=24,513$, $L=94$, $m \approx 25$, 15 outer iterations, 3 inner E-steps).

Converged hyperparameters.

10 NUFFT budget analysis

NUFFTs are $O(n)$ and dominate wall-clock time. Toeplitz matvecs are $O(m \log m)$ and essentially free for $m=25$.

	ℓ_g	σ_g^2	ℓ_h	σ_h^2	r
B: Block CD	0.085	4.04	0.105	2.10	2.84
C: Schur CG	0.196	1.57	0.072	1.17	2.84
D: Direct	0.313	0.95	0.083	1.85	2.84

Table 2: Learned hyperparameters after 15 iterations. Despite different ℓ_g/σ_g^2 tradeoffs, all approaches reach similar MAE (≈ 27.4), indicating that the posterior predictive is robust to this identifiability.

	B: Block CD	C: Schur CG	D: Direct
E-step (per inner)	$10(1+L) = 950$	$2(1+L) = 190$	$2(1+L) = 190$
sigma_diag (per inner)	$2(1+L) = 190$	$2(1+L) = 190$	$2(1+L) = 190$
Inner iters $\times 3$	3420	1140	1140
M-step	0	0	0
Total per outer	3420	1140	1140

Table 3: NUFFT calls per outer iteration. Approaches C and D have identical NUFFT budgets; D is faster because it replaces CG with direct solves.

11 Recommendations

- $d=1$, **moderate** m : Use Approach D (Direct Schur). $8.4\times$ faster than joint, no CG, exact traces. Works whenever $m \lesssim 200$.
- $d \geq 2$ or **very small lengthscale**: Use Approach C (Schur CG). Same NUFFT budget as Direct but avoids dense $m^d \times m^d$ factorization. The nested CG adds Toeplitz overhead but scales gracefully.
- **Approach B** (Block CD) is superseded: it uses $5\times$ more NUFFTs than Schur-based approaches and introduces a CD approximation error in the joint posterior.